

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

GCSE MATHEMATICS

F

Foundation Tier Paper 3 Calculator

Tuesday 12 June 2018

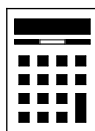
Morning

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a calculator
- mathematical instruments.



Instructions

- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 80.
- You may ask for more answer paper, graph paper and tracing paper. These must be tagged securely to this answer book.

For Examiner's Use

Pages	Mark
2–3	
4–5	
6–7	
8–9	
10–11	
12–13	
14–15	
16–17	
18–19	
20–21	
22–23	
24–25	
26–27	
TOTAL	

Advice

- In all calculations, show clearly how you work out your answer.



J U N 1 8 8 3 0 0 3 F 0 1

Answer **all** questions in the spaces provided

Do not write
outside the
box

- 1 Circle the value of the digit 7 in 9.17

[1 mark]

$$\frac{1}{70}$$

$$\frac{1}{7}$$

$$\frac{7}{10}$$

$$\frac{7}{100}$$

- 2 Solve $3x = 2$
Circle your answer.

[1 mark]

$$x = -1$$

$$x = \frac{2}{3}$$

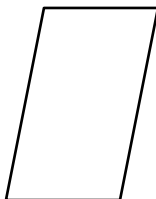
$$x = \frac{3}{2}$$

$$x = 6$$

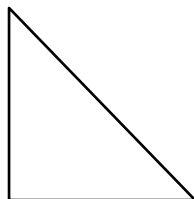
- 3 Which of these shapes has **no** lines of symmetry?
Circle the correct letter.

[1 mark]

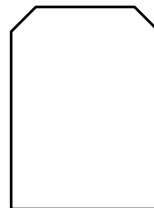
A



B



C



D



4 Circle the shortest length.

[1 mark]

1200 cm

0.13 km

110 m

140 000 mm

5 (a) Shade $\frac{2}{5}$ of this grid.

[1 mark]

5 (b) Shade 10% of this grid.

[1 mark]



6 Saj wants to go to all 19 home games at a football club.

For each game, a ticket costs £28

A season ticket

costs £379

and

gives entry to all 19 home games.

In total, how much does Saj save by buying a season ticket?

[3 marks]

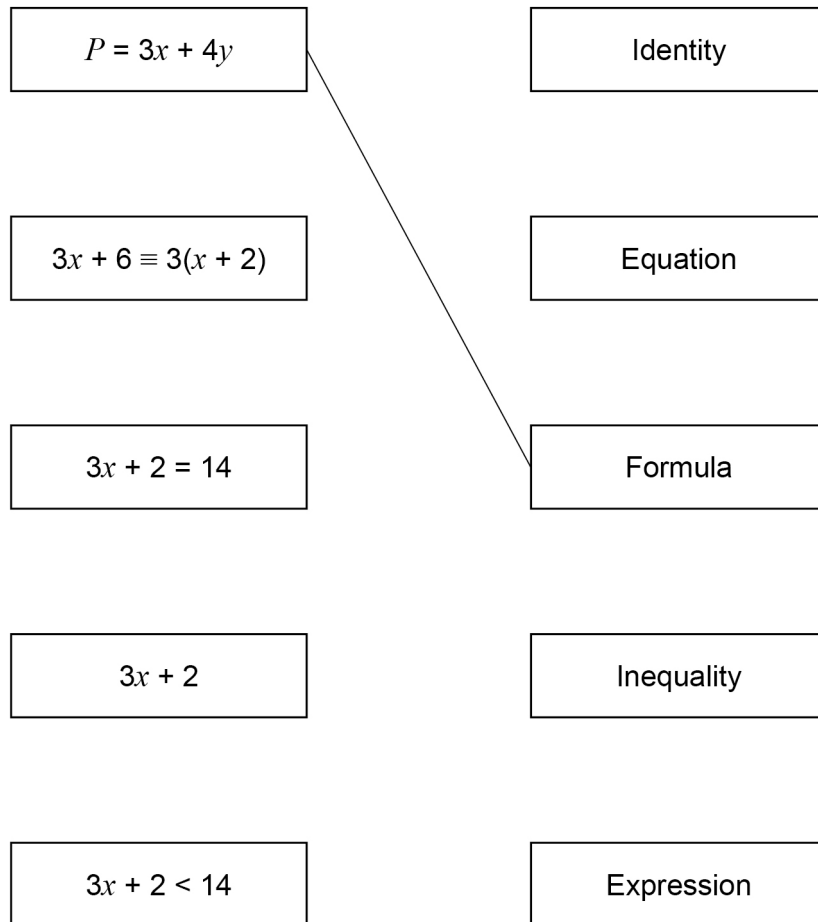
Answer £ _____



7

Link the algebra to the correct description.

One has been done for you.

[3 marks]**Turn over for the next question****Turn over ►**

8

Jim has six banknotes.

The value of each note is £5 or £10 or £20

He **can** make £20 with three notes.

He **can** make £55 with four notes.

He **cannot** make £25 with three notes.

He **cannot** make £25 with four notes.

List the six notes.

[2 marks]

£ _____ £ _____ £ _____

£ _____ £ _____ £ _____



- 9** A music app has a shuffle play function.
This means that songs are played in a random order **without repeat**.

- 9 (a)** Ruth puts 10 songs on shuffle play.
One of them is her favourite song.
Write down the probability that her favourite song plays first.

[1 mark]

Answer _____

- 9 (b)** Ted puts songs A, B and C on shuffle play.
List all the possible orders of songs A, B and C.
One has been done for you.

[2 marks]

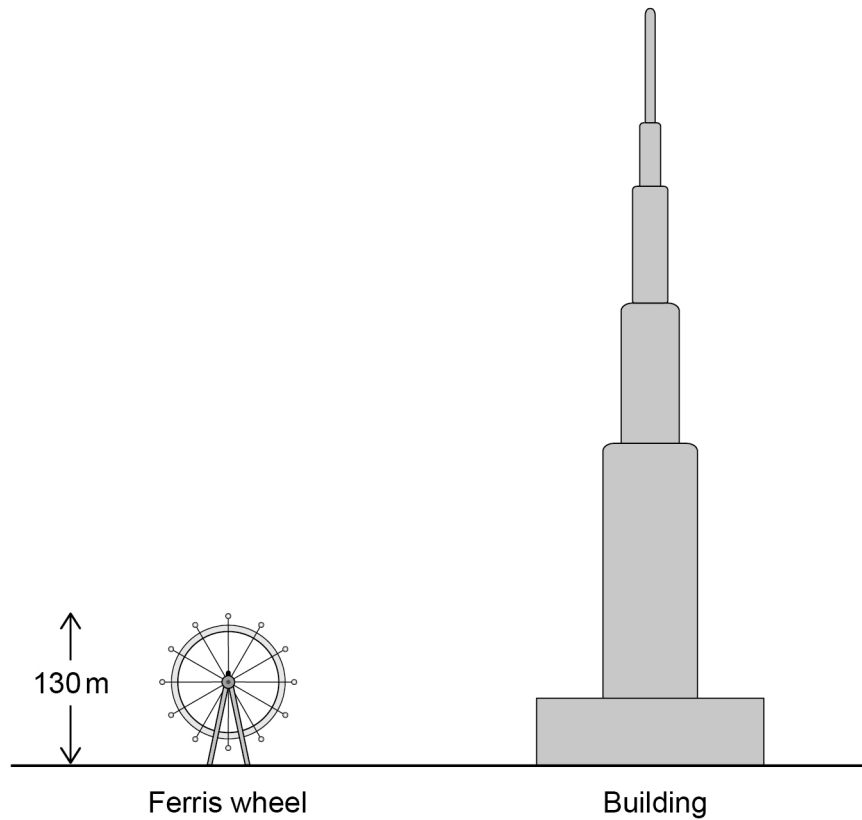
A B C

Turn over for the next question



10

Here is a scale drawing.



The Ferris wheel has a height of 130 m

Work out the height of the building.

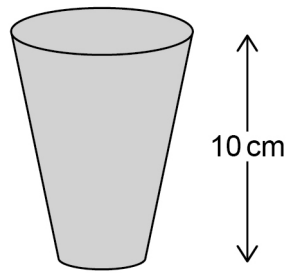
[3 marks]

Answer _____ m

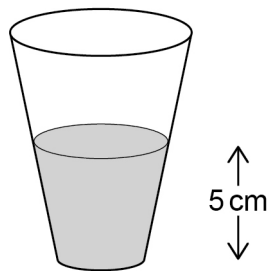


11

Jo has a full cup of coffee.



She drinks some of it.



She says,

“Half of the coffee is still in the cup, because 5 cm is half of 10 cm”

Is she correct?

Tick a box.

☐

Yes

☐

No

Give a reason for your answer.

[1 mark]



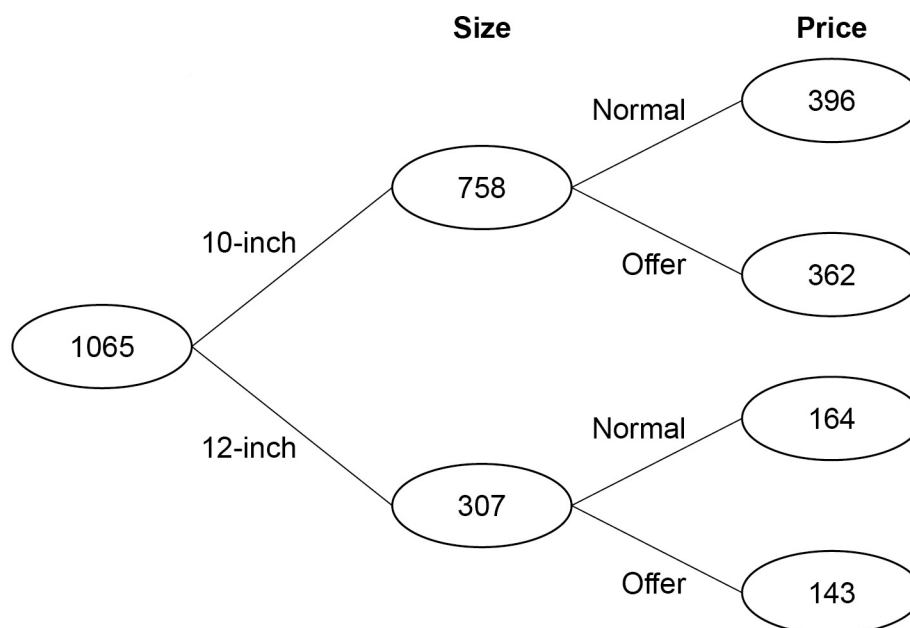
12

A takeaway sells 10-inch pizzas and 12-inch pizzas.

Here is some information about the numbers sold in two weeks.

Week 1

10-inch	512
12-inch	231
Total	743

Week 2

12 (a) In each week a proportion of the pizzas sold were 10-inch.

In which week was this proportion greater?

Show working to support your answer.

[2 marks]

Answer _____



12 (b)

	Normal price	Offer price
10-inch	£3.74 profit	51p loss
12-inch	£5.29 profit	4p loss

In week 1 the total profit was £1895.55

At the end of week 1 the takeaway spent £175 on adverts.

Was the **increase** in profit in week 2 more than the cost of the adverts?

You **must** show your working.

[4 marks]

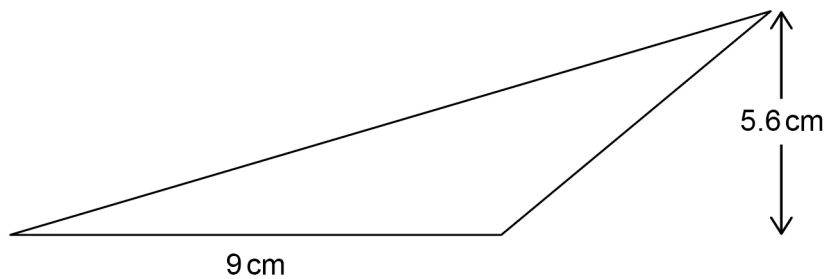
This image shows a single page of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page, leaving small margins at the top and bottom. There are no vertical margin lines, and the page is completely blank except for the lines themselves.

Answer

- 13** A car travels 3.5 miles in 5 minutes.
Work out the average speed in miles per hour. **[3 marks]**

Answer _____ mph

- 14** A triangle has base 9 cm and perpendicular height 5.6 cm Not drawn accurately



- Work out the area of the triangle. **[2 marks]**

Answer _____ cm^2

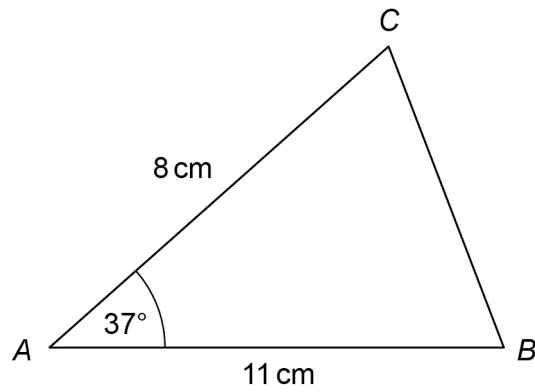


- 15** Four positive whole numbers add up to 36
One of the numbers is a multiple of 7
The other three numbers are equal.
Work out the result when the four numbers are multiplied. **[3 marks]**

Answer _____



16

A sketch of triangle ABC is shown.Not drawn
accuratelyIn the space below, complete an accurate drawing of triangle ABC .**[2 marks]**

A ————— B



- 17** Simplify $7x - (3x - 2x)$
Circle your answer.

[1 mark]

$7x - 1$

$2x$

$6x$

$8x$

- 18** A competition
took place in 1983
takes place every six years.
Circle the year in which it will also take place.

[1 mark]

2083

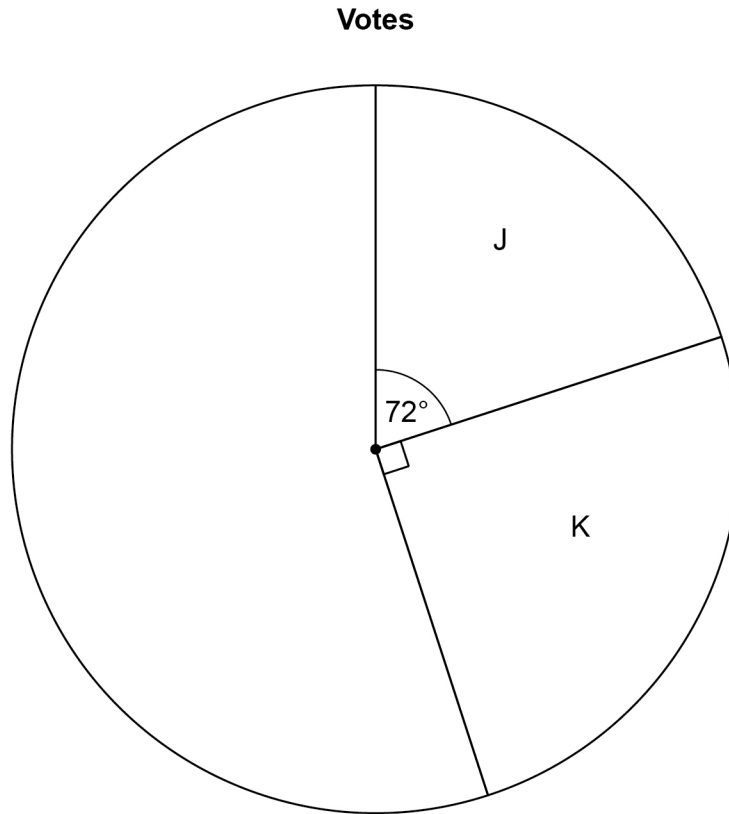
2036

2049

2023

Turn over for the next question**Turn over ►**

- 19** In an election there were four candidates, J, K, L and M.
Fran is drawing a pie chart to show the results.
The sectors for J and K have been drawn.



- 19 (a)** Twice as many people voted for L as voted for M.
Complete the pie chart.

[3 marks]

19 (b) Altogether, 16 200 people voted.

How many voted for J?

[2 marks]

Answer _____

20 The probability that A is the outcome of an experiment is 0.2

Circle the probability that A is **not** the outcome.

[1 mark]

0

0.2

0.5

0.8

21 Rearrange $e = 2f$ to make f the subject.

Circle your answer.

[1 mark]

$$f = 2e$$

$$f = \frac{2}{e}$$

$$f = e - 2$$

$$f = \frac{e}{2}$$

Turn over for the next question



22

After the first two terms, each term is half the sum of the previous two terms

22 (a)

2 10 6

Show that the 6th term is the first one that is **not** a whole number.

[3 marks]

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper has a slight shadow on the right side, suggesting it's resting on a surface.

22 (b) A different sequence follows the same rule.

The 1st term is 4

The 3rd term is 9.5

4 9.5

Work out the 2nd term.

[3 marks]

Answer _____

Turn over for the next question



23

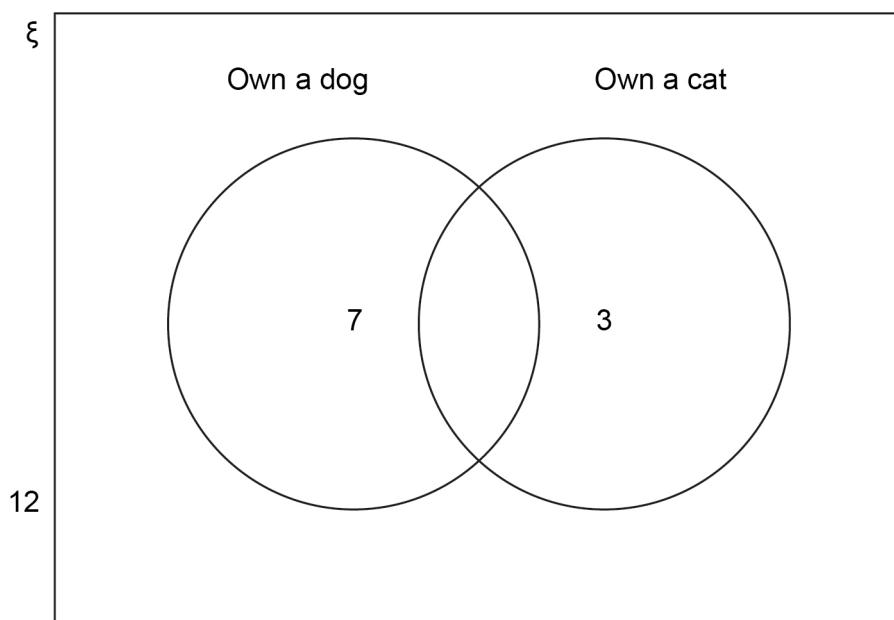
In a group of 20 people

7 own a dog

3 own a cat

12 do not own a dog or a cat.

Aidan shows this information on a Venn diagram.

Make **two** criticisms of his Venn diagram.**[2 marks]**

Criticism 1 _____

Criticism 2 _____



24 a is a common factor of 72 and 120 b is a common multiple of 6 and 9Work out the highest possible value of $\frac{a}{b}$ **[4 marks]**

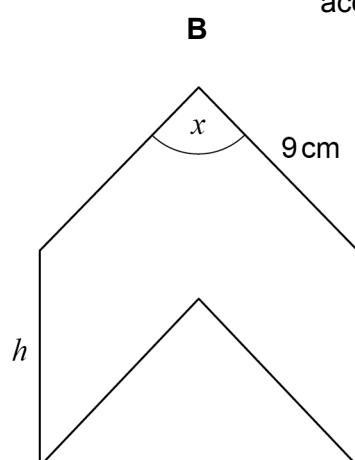
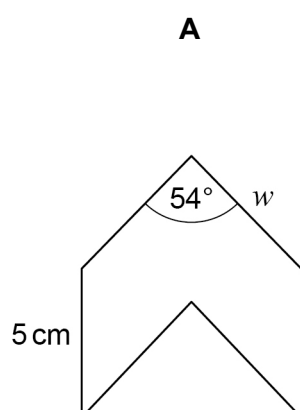
Answer _____

Turn over for the next question

25

A and B are similar shapes.

B is an enlargement of A with scale factor 1.5

Not drawn
accuratelyWork out the values of x , h and w .**[3 marks]**

 $x =$ _____ degrees $h =$ _____ cm $w =$ _____ cm

26

Investment A Save £150 per month for 2 years.
2.5% interest is added to the total amount saved.

Investment B Invest £3500
Compound interest is added at 3% per year.

After 2 years, how much **more** is investment B worth than investment A?

[4 marks]

Answer £ _____

Turn over for the next question



- 27 (a)** Show that the lines $y = 3x + 7$ and $2y - 6x = 8$ are parallel.
Do **not** use a graphical method.

[3 marks]

- 27 (b)** Is the point $(-5, -6)$ above, below or on the line $y = 3x + 7$?
Tick **one** box.

☐

Above

☐

Below

☐

On the line

You **must** show your working.
Do **not** use a graphical method.

[2 marks]



28 The cost of a ticket increases by 10% to £19.25

Work out the original cost.

[3 marks]

Answer £ _____

Turn over for the next question



29 The n th term of a sequence is $12n - 5$

Work out the numbers in the sequence that
have two digits
and
are **not** prime.

[3 marks]

Answer _____



30 $\mathbf{a} = \begin{pmatrix} 6 \\ -10 \end{pmatrix}$ $\mathbf{b} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$ $\mathbf{c} = \begin{pmatrix} -4 \\ 7 \end{pmatrix}$

30 (a) Work out $\mathbf{a} + \mathbf{b} + \mathbf{c}$

[2 marks]

Answer

 $\left(\right)$

30 (b) Show that $\mathbf{a} + 2\mathbf{c} = k\mathbf{b}$, where k is an integer.

[2 marks]

END OF QUESTIONS

